



RAO-BLACKWELL THEOREM

The Rao-Blackwell Theorem tells us that the efficiency of an unbiased estimator can be improved if we take an unbiased estimator which is also a function of a sufficient statistic.

STATEMENT

Let $x_1, x_2, \dots, x_n$ be a random sample from a distⁿ $f(x; \theta)$ such that

$Y_1 = t(x_1, x_2, \dots, x_n)$ is a sufficient statistic of parameter θ and $Y_2 = h(x_1, x_2, \dots, x_n)$ be an other unbiased statistic i.e. $E(Y_2) = \theta$. If conditional function $E[Y_2/Y_1 = y_1] = \phi(y_1)$ is a function of Y_1 and is independent of θ , then

$$(i) \quad E[\phi(y_1)] = \theta \quad \text{and}$$

$$(ii) \quad \text{Var}[\phi(y_1)] < \text{Var}(Y_2)$$

Proof: By definition

$$E[\phi(y_1)] = \int_{-\infty}^{\infty} \phi(y_1) g(y_1) dy_1$$

$$E(x) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$\text{as } \phi(y_1) = E(Y_2/Y_1 = y_1) = \int_{-\infty}^{\infty} y_2 f(y_2/y_1) dy_2$$

(2)



$$\text{hence } E[\phi(y_1)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} y_2 f(y_2/y_1) dy_2 \right] g(y_1) dy_1$$

As we know $f(y_2/y_1) = \frac{f(y_1, y_2)}{g(y_1)} = \frac{\text{Joint dist}^n}{\text{Marginal dist}^n}$
(from conditional probability)

$$E[\phi(y_1)] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} y_2 \frac{f(y_1, y_2)}{g(y_1)} dy_2 \right] g(y_1) dy_1$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y_2 f(y_1, y_2) dy_1 dy_2$$

$$= \int_{-\infty}^{\infty} y_2 \left[\int_{-\infty}^{\infty} f(y_1, y_2) dy_1 \right] dy_2$$

$$= \int_{-\infty}^{\infty} y_2 f(y_2) dy_2 \quad \left| \begin{array}{l} g(y_2) = \int f(y_1, y_2) dy_1 \\ \text{marginal pdf} \end{array} \right.$$

$$E[\phi(y_1)] = E(y_2) = \theta \quad , \quad \text{proved part (i)}$$

Now part ii

$$\text{Let } \text{Var}[\phi(y_1)] = \sigma_c^2 \quad \text{and}$$

$$\text{Var}(y_2) = \sigma_o^2$$

We have to prove that $\sigma_c^2 < \sigma_o^2$, the
Variance of y_2 can be written as

$$\sigma_o^2 = E[y_2 - E(y_2)]^2 = E[y_2 - \theta]^2$$

subtracting and adding $\phi(y_1)$



$$\sigma_0^2 = E \left[y_2 - \phi(y_1) + \phi(y_1) - \theta \right]^2$$

$$= E \left[\{ y_2 - \phi(y_1) \} + \{ \phi(y_1) - \theta \} \right]^2$$

expanding the square

$$= E \left[\{ y_2 - \phi(y_1) \}^2 + \{ \phi(y_1) - \theta \}^2 + 2 \{ y_2 - \phi(y_1) \} \{ \phi(y_1) - \theta \} \right]$$

applying the expectation

$$\sigma_0^2 = E \{ y_2 - \phi(y_1) \}^2 + E \{ \phi(y_1) - \theta \}^2 + 2 E \{ \{ y_2 - \phi(y_1) \} \{ \phi(y_1) - \theta \} \}$$

(A)

Consider $E \{ \{ y_2 - \phi(y_1) \} \{ \phi(y_1) - \theta \} \}$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y_2 - \phi(y_1)) (\phi(y_1) - \theta) g(y_1, y_2) dy_1 dy_2$$

$$\text{As } g(y_2/y_1) = \frac{g(y_1, y_2)}{g(y_1)}$$

$$\Rightarrow g(y_1, y_2) = g(y_2/y_1) g(y_1)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \{ y_2 - \phi(y_1) \} \{ \phi(y_1) - \theta \} g(y_2/y_1) g(y_1) dy_1 dy_2$$

$$= \int_{-\infty}^{\infty} \{ \phi(y_1) - \theta \} \left[\int_{-\infty}^{\infty} \{ y_2 - \phi(y_1) \} g(y_2/y_1) dy_2 \right] g(y_1) dy_1$$

$$= \int_{-\infty}^{\infty} \{ \phi(y_1) - \theta \} \left[\int_{-\infty}^{\infty} y_2 g(y_2/y_1) dy_2 - \int_{-\infty}^{\infty} \phi(y_1) g(y_2/y_1) dy_2 \right] g(y_1) dy_1$$

$$= \int_{-\infty}^{\infty} \{ \phi(y_1) - \theta \} \left[E(y_2/y_1) - \phi(y_1) \int_{-\infty}^{\infty} g(y_2/y_1) dy_2 \right] g(y_1) dy_1$$

(4)



$$= \int_{-\infty}^{\infty} \{ \phi(y_1) - \theta \} [\phi(y_1) - \phi(y_1 | 1)] g(y_1) dy_1$$

$$= \int_{-\infty}^{\infty} \{ \phi(y_1) - \theta \} [\phi(y_1) - \phi(y_1)] g(y_1) dy_1$$

= 0, now the equation (A) becomes

$$\sigma_o^2 = E \{ Y_2 - \phi(Y_1) \}^2 + E \{ \phi(Y_1) - \theta \}^2$$

$$\sigma_o^2 = E \{ Y_2 - \phi(Y_1) \}^2 + \sigma_c^2$$

It shows that σ_o^2 is greater than σ_c^2 by a positive quantity $E \{ Y_2 - \phi(Y_1) \}^2$ so

$$\sigma_o^2 > \sigma_c^2 \quad \text{or}$$

$$\sigma_c^2 < \sigma_o^2$$

$$\text{Var} [E(Y_2 / Y_1 = y_1)] < \text{Var}(Y_2) \quad \text{or}$$

$$\text{Var} [\phi(y_1)] < \text{Var}(Y_2), \text{ which}$$

completes the proof of the part II.